A Study on Behaviour Pattern and Dormancy Risk of Unclaimed Deposits in Indian Banks (2015-2024)

Submitted to

MANIPAL SCHOOL OF COMMERCE AND ECONOMICS

MANIPAL ACADEMY OF HIGHER EDUCATION

In Partial Fulfilment of the Requirements of the

Awards of the Degree

of

M Sc. Business Analytics

By

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DECLARATION

I hereby declare that this project report titled "A Study on Behaviour Pattern and Dormancy Risk of Unclaimed Deposits in Indian Banks (2015–2024)" submitted by me to Manipal Academy of Higher Education in partial fulfilment of the requirements of the Degree of M.Sc. Business Analytics is a record of Bonafide work carried out by me under the guidance of Dr. Everil Jacklin Fernandes, Manipal School of Commerce and Economics, Manipal Academy of Higher Education, Manipal.

This report is the outcome of my work with Dr. Everil Jacklin Fernandes, and the contents of this report have not previously formed the basis for the award of any Degree, Diploma or such other similar title. I further declare that the sources and references used in this research have been duly acknowledged and cited.

Date: 30-04-2026

Place: Manipal

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CERTIFICATE

This is to certify that project report titled ***"A study on Behaviour pattern and Dormancy Risk of Unclaimed Deposits in Indian Banks (2015-2024)"*** is submitted to Manipal school of commerce and Economics, Manipal Academy of Higher Education (MAHE), Manipal, towards partial fulfillment of the requirements for the award of ***M.SC (BUSINESS ANALYTICS)*** degree.

Disha Bhoja Poojari, bearing Reg No: ***242626039,*** has worked under my supervision and guidance. No part of this report was earlier submitted for the award of any Degree, Diploma, Fellowship or any other similar title or prizes. The work has not been published in any journal or magazine.

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ACKNOWLEDGEMENT

I extend my deepest gratitude to those who contributed to the success of this project. First, I would like to thank Manipal Academy of Higher Education, especially the Manipal School of Commerce and Economics, for providing the resources and support necessary to carry out this research.

I would like to thank Dr. Everil Jacklin Fernandes, whose insights and expertise were invaluable throughout the duration of this project. Her guidance and academic rigour greatly enhanced the quality of this work and my understanding of the complex issues addressed.

I would like to acknowledge the faculty members of Department of Commerce for their encouragement and for fostering a research-oriented environment. Also Special thanks to my colleagues and peers who engaged in valuable discussions during the research process, helping me refine my analytical approach and strengthen my conclusions.

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LIST OF ABBREVIATIONS AND SYMBOLS

ABBREVIATIONS:

- CA Current Account
- SA Savings Account
- FD Fixed Deposit
- OD Others/Other Deposits
- PSB Public Sector Bank
- RRB Regional Rural Bank
- SFB Small Finance Bank
- RBI Reserve Bank of India
- DICGC Deposit Insurance and Credit Guarantee Corporation
- PMJDY Pradhan Mantri Jan Dhan Yojana
- HHI Herfindahl–Hirschman Index
- EDA Exploratory Data Analysis
- KPI Key Performance Indicator
- KNN K-Nearest Neighbors
- GMM Gaussian Mixture Model

SYMBOLS:

- τ Kendall's Tau (trend test statistic)
- ρ Spearman's Rho (correlation coefficient)
- H Kruskal-Wallis test statistic
- U Mann-Whitney U test statistic
- p Probability value (significance)
- α Significance level
- η^2 Eta squared (effect size)
- μ Mean
- σ Standard deviation
- Σ Summation
- $>$ Greater than
- $<$ Less than

- ≤ Less than or equal to
- ≥ Greater than or equal to
- % Percentage
- ₹ Indian Rupee
- Cr Crore (10 million)

ABSTRACT

In India, unclaimed or dormant deposits are an ongoing problem in the banking sector, which has seen an increase from ₹5,161 Crore in 2015 to ₹62,234 Crore by 2024 (12 times). These statistics clearly point to areas of concern about systematic deposit management. This study looks at behavioural trends and risk since the last decade (2015 to 2024) across 175 banks with sophisticated data analysis techniques employed to determine concentration, growth trajectory and risk heterogeneity.

The study uses different methodologies: (1) K-Means clustering with K=3 to identify three separate behaviour patterns for dormancy based on growth rates and lag deposits; (2) Herfindahl-Hirschman Index (HHI) analysis of the concentration of dormancy for account types (Current Account, Savings Account, Fixed Deposit and Other); and (3) composite risk score that incorporates growth, size and concentration metrics. Findings for the five different hypothesis tests (Kruskal-Wallis, Mann-Kendall, Mann-Whitney U, Spearman correlation and cluster comparison) are statistically significant for $p < 0.05$.

Main Findings: Inactivity has led unclaimed deposits to increase dramatically (approximately double from 2021) largely due to disuse of savings/checking accounts and unclaimed fixed deposit accounts. By the middle of 2025 it is evident that public sector banks (PSBs) have largest volume of unclaimed deposits with thousands of crores in unclaimed deposit accounts.

There are fundamental differences in the way banks have amassed their deposits over time. Results of clustering analysis (K-means) show banks can be grouped into identifiable clusters that have similar donation behaviours, i.e. high and low growth or high risk and low risk. There are significant variances in the risk profile of these clusters and each cluster represents different levels of risk with respect to the underlying bank.

Using lagged deposit data, recent growth rate data and each bank's clustering membership we combine these variables to create a Risk Score. Banks classified as high risk based on their Risk score (e.g. large PSB's and some rapidly growing private sector banks) deserve immediate supervisory attention from regulators; likewise, banks identified as high risk in this analysis will continue to grow at a higher rate relative to their peers.

The analysis above improves our understanding of banking institution dormancy through analysis of the patterns of dormant deposits, rather than merely aggregate totals. These findings suggest regulators and banking institutions should consider expanding their customer outreach efforts, increasing visibility of unclaimed deposits including providing unclaimed deposit reporting to the general public and making UDGAM Portal available to depositors for reclaiming their funds. This study provides regulatory authorities with important information to better understand and manage through easily calculated, quantified dormancy risk profiles; validate multi-factor risk assessment frameworks; and present data driven suggestions for the Reserve Bank of India's supervision. Limitations include concerns surrounding data quality after 2020 and restricting its analysis and findings to the domestic banking system.

Keywords: Dormant Deposits, Bank Risk, Cluster Analysis, Concentration Risk, HHI, Deposit Management, K-Means Clustering, Risk Scoring, Indian Banking

1. INTRODUCTION

1.1 Background and Context

The Banking Industry in India has deposits of trillions upon trillions of rupees and is at the centre of all financial activity occurring in the economy, but a substantial segment is in what are referred to as "unclaimed" or "dormant" deposits, having had no activity for a period of time, both representing a Regulatory challenge, and a record of the effectiveness of the banking sector as a whole in managing deposits. As defined in the rules established by the Reserve Bank of India, as dormant deposits are deposit accounts having had no activity for a period of time, typically two years.

The increase in outstanding dormant deposits in the Indian banking industry has been substantial, increasing from ₹5,161 Crore in 2015, to ₹62,234 Crore in 2024. This represents an increase of 12x over the course of a decade, and indicates that problems, that have persisted over that period of time in managing deposit life cycle, in customer engagement and in financial inclusion at the system level, have not improved at all, as the number of dormant deposits has continued to increase.

While dormancy is seen as a technical banking issue, its implications are serious and multi-faceted with regards to customer trust, regulatory compliance and system wide stability. Dormant deposits are growing at rates that significantly exceed the growth of nominal GDP and Inflation indicating that the dramatic rise in dormant deposits is not merely a by-product of the expansion of the overall deposit base, but are a reflection of serious and long-standing issues confronting banks in managing inactive accounts on a system-wide basis. Financial inclusion initiatives such as Pradhan Mantri Jan Dhan Yojana (PMJDY) has expanded banking access dramatically, yet current growth in dormancy suggests that account opening alone does not ensure sustained customer engagement or active financial participation.

Although there is evidence that dormant deposits are increasing, our empirical understanding of how they behave and what leads to this behaviour is still limited. In India, there is large heterogeneity in the banking sector with many different kinds of banks including Public Sector Banks, Private Banks, Regional Rural Banks, Cooperative Banks and Small Finance Banks; however, no systematic study of dormancy patterns across the different categories exists.

1.2 Problem Statement

There are several gaps in the existing research on dormancy in the Indian banking system:

- There has been no quantification of dormancy concentration patterns across different types of accounts within banks which limits our ability to assess potential vulnerabilities to account type specific risks.
- There has been no differentiation in dormancy profiles of the various categories of banks which hinders targeted regulatory Intervention or policy formulation.
- There has been little analysis of the temporal nature of dormancy trends and the impact of major policy interventions and macro events (e.g. demonetization; COVID-19) in shaping these trends.
- There are no established multi-factor risk frameworks for assessing dormancy that consider growth trajectories, the scale of deposits and concentration at the same time.
- Most importantly, there has been a lack of investigation into whether the growth of dormancy is attributable to structural issues within the banking system or if it reflects temporary phenomena related to specific events or policies.

This research will fill these gaps by providing a comprehensive quantitative analysis of dormancy behaviour across 175 banks using clustering approaches, concentration metrics, and statistical hypothesis testing.

1.3 Research Questions

This study is guided by the following research questions:

1. How does dormancy evolve across Indian banks from 2015 to 2024, and what are the underlying growth trends?

2. Can distinct dormancy behaviour profiles be identified across banks, and if so, what characteristics define each profile?
3. How concentrated is dormancy across account types within individual banks, and which institutions face elevated concentration risk?
4. How do dormancy risk levels vary across different bank types and institutional sizes?
5. What are the drivers of dormancy risk, and how can a composite risk framework effectively measure dormancy exposure?
6. What are the policy implications for regulatory monitoring and intervention strategies based on identified dormancy patterns?

1.4 Research Objectives

In response to these research questions, the following objectives are pursued:

1. To analyse the trend and growth of unclaimed (dormant) bank deposits across different banks over time.

2. To identify patterns in dormancy behaviour among banks using clustering techniques based on lagged deposit values and growth rates.

3. To measure the concentration of unclaimed deposits across different account types (Current, Savings, Fixed Deposits, and Others) using the Herfindahl–Hirschman Index (HHI).

4. To develop a composite dormancy risk score that ranks banks based on factors such as deposit growth, lagged deposit values, and cluster classification.

5. To compare dormancy risk levels across different categories of banks and identify institutions that show higher vulnerability to deposit dormancy.

6. To provide data-driven insights that can help regulators or financial institutions better understand and monitor the risk associated with unclaimed deposits.

1.5 Significance of the Study

The paper has provided a number of contributions to the literature on dormancy and banking regulation:

Academic: Using a rigorous statistical methodology (5 hypothesis tests, multi-factor risk scoring, clustering validation), the paper examines an under-studied part of banking system behaviour and has enhanced the empirical understanding of dormancy as a systematic phenomenon rather than an isolated operational issue.

Policy: The research findings will be used to inform the regulation and supervisory framework of the RBI. This will allow for differentiated approaches to dormancy management by bank category, as well as identifying which banks should be subject to greater oversight.

Practical: Banks can use dormancy profiles and risk scores to enhance customer engagement, better manage deposit lifecycles, and improve operational efficiencies.

Systemic Stability: Understanding the concentration and growth of dormancy will increase the ability to conduct macro-prudential oversight, as a concentration of dormancies can indicate potential systemic issues with the management of deposits with implications for the stability of the financial system.

1.6 Scope And Boundaries

- The research was conducted over a ten-year period (2015-2024), which included pre-COVID, COVID, and post-COVID time periods.
- 175 unique banks were analysed to reflect the full spectrum of the Indian Banking System based on the 9 individual bank types (1,228 bank-year observations after data cleaning).
- Deposit accounts were analysed based on 4 different account types (Current Accounts, Savings Accounts, Fixed Deposit, Other) to review concentration patterns.
- The analysis involved Clustering methods (K-Means), Concentration Analysis (HHI), Hypothesis Testing (5 statistical tests) and Risk Scoring (3-factor model).
- The Study focuses on dormancy as a passive state (no transactions), not explicitly accounting for customer demographics, branch characteristics, or policy-specific campaigns as explanatory variables.

2. REVIEW OF LITERATURE

Unclaimed deposits and dormant accounts have become a key issue in today's banking systems because they affect financial inclusion, how smoothly banks operate, and how well they manage regulatory risks. As banking becomes more available worldwide, the number of inactive accounts has gone up. This means there's a growing need for better ways to track and analyse those dormant balances. Most of the existing research focuses on three main topics: how unclaimed deposits change over time, issues around financial inclusion and inactive accounts, and how banks are using machine learning for their data analysis.

2.1 Unclaimed Deposita and Dormant Accounts in Banking

Unclaimed deposits and dormant accounts in banking refer to money or accounts that haven't been accessed or claimed by their owners for a long time. These funds often sit idle in bank records because the account holder has either forgotten about them or lost touch with the bank. Banks usually follow specific rules to handle these accounts, like trying to notify the owner or transferring the money to a government agency if no one comes forward after a certain period. It's important for people to keep track of their accounts and update their contact information to avoid losing access to their money. A number of studies have looked into how dormant accounts affect operations and finances. Research on dormant financial instruments shows that inactive balances can impact how banks manage their liquidity and can also create governance risks, especially if their monitoring systems aren't strong (Okeke & Eze, 2020).Studies looking at unclaimed deposits in Indian banks point out that idle money shows financial resources that aren't being used well and might also indicate that banks aren't connecting with customers as effectively as they could (Journal Press India, 2023).Regulatory analyses, like the RBI guidelines on inoperative accounts, explain that dormant balances mainly come from accounts that haven't been used for a long time. This usually happens because people move away, don't adopt digital banking, or simply aren't aware of how their accounts work. Research from around the world also looks at the rules and issues that come up with dormant balances. Legal studies about managing dormant accounts in places like the UAE and Indonesia show gaps in how these accounts are monitored. They also point out the need for risk-based supervision to handle these issues better (ResearchGate, 2024). These studies lay the groundwork for why keeping an eye on dormant deposits is becoming more and more important for policy decisions.

2.2 Financial Inclusion and Account Inactivity

The growth of financial inclusion has significantly led to a big rise in bank accounts around the world, but many of these accounts aren't active anymore. Research on global financial inclusion shows that many people who have accounts don't use them much, leaving balances that stay inactive even though more people have access to banking services (Demirgüç-Kunt et al.). Research on inclusion programs in India shows that things like low digital skills, difficulties with rural banking, and people moving for work all play a role in accounts becoming inactive. Research on dormant accounts in financial inclusion programs suggests that inactive balances can make these policies less effective. Just opening accounts to meet policy goals doesn't always lead to ongoing financial activity or involvement (Banking Law Research Studies, 2022). These findings show why it's important to look at behavioural patterns instead of just focusing on total deposit numbers.

2.3 Customer behaviour, financial inclusion and dormancy drivers

Some literature focuses on customer behaviour and the determinants of account inactivity. Global Financial inclusion studies led by Klapper and colleagues show that demographic factors, income levels, financial literacy and product design all influence whether individuals actively use their accounts or allow them to lapse into dormancy. Banking laws and policy work on dormant PMJDY accounts in India links dormancy to low balances, irregular income flows and low understanding of account features, suggesting that inclusion incentives generate a large pool of inactive users or account holders unless it is complemented by a sustained engagement. These findings suggests that dormancy is an outcome of behavioural pattern that is shaped by both customer characteristics and institutional practices.

Closely related studies on customer retention and dormant-account reactivation propose targeted outreach strategies, better communication and incentive-based schemes to reengage inactive customers. Research on "early customer inactivity" or churn prediction uses detailed transaction histories and behavioural features to identify signals of impending inactivity or relationship termination well before accounts are formally declared dormant. However, most of this work is conducted at the customer or account level within a single institution and focuses on marketing or retention outcomes, rather than on aggregating unclaimed balances and dormancy risk at the bank level across an entire banking system.

2.4 Behavioural Analytics and Machine Learning in Banking

Behavioural analytics and machine learning in banking are changing how banks understand their customers. By studying customer actions and patterns, banks can spot fraud faster, offer better services, and make smarter decisions. Machine learning helps banks analyse large amounts of data quickly, which improves accuracy and personalizes banking experiences. This combination is becoming important for banks to stay competitive and meet customer needs more effectively. Machine learning has become more common in financial analytics, especially when it comes to assessing credit risk, spotting fraud, and understanding customer behaviour. Reviews of machine learning in banking show that regression models and ensemble methods do a good job of spotting behavioural trends in financial data (Leo et al.,2019).

Research into predicting inactive banking customers shows that looking at things like how often they make transactions and their past balances can help spot dormant accounts before they become a bigger issue (ScienceDirect, 2023). Even with these advancements, most machine learning research still looks at customer-level data instead of how institutions behave over time. Research on predicting customer inactivity often focuses on behavioural features to spot risk patterns, but there’s not much work that applies these kinds of analyses to bank-level dormant deposit data. Studies on machine learning in financial risk management show the growing role of data-driven decision frameworks, though most focus on credit or market risk and don’t really cover operational dormancy behaviour (Bussmann et al.).

2.5 Research gap

Most of the earlier studies offer useful information about why accounts become dormant and what happens afterward, but they mostly focus on simple statistics, rules, or look at things mainly from the customer's point of view. Not many studies use machine learning techniques to look at the behavioural patterns of unclaimed deposits in institutions. There isn’t much research that combines trend analysis, behavioural clustering, and composite risk scoring to profile dormancy risk across banks.

Contribution of the present study

This research aims to fill the gap by offering a behaviour driven analytical framework that uses machine learning to study unclaimed deposits in Indian banks. The study goes beyond just describing data by using temporal trend analysis, checking behaviour with regression, grouping similar cases through clustering, and creating a Dormancy Risk Score to classify banks into risk categories, extending existing risk-management applications of machine learning to the specific context of dormant accounts and unclaimed deposits in India. This approach helps build a risk profile that can guide decisions. This framework gives a clearer view of risks tied to growth and long-term dormancy, which could be useful for regulators and policy makers when keeping an eye on things and making decisions.

3. CONCEPTUAL FRAMEWORK

3.1 Theoretical Foundations

There are three theoretical foundations of dormancy:

1. The Behavioural Finance Theory provides an understanding of dormant accounts to occur due to the individual's decision-making as influenced by financial literacy, product characteristics or design and life situations/disruptions
2. Systems Theory which views Banking as a multi-dimensional interconnected system where concentration of dormancy at certain institutions or in certain account types creates systemic risk.
3. Risk Management Frameworks which define dormancy as both an operational and reputational risk and regulatory risk which has to be measured, monitored and managed systematically.

Dormancy can be viewed from a risk perspective in the following three ways:

- Temporal Dimension (Development): How rapidly dormancy is developing.
- Scale Dimension (Dormancy Size): the extent of dormant funds accumulated.
- Structural Dimension (Clustering): The pattern of behaviour of the institutions involved.

There is a complex inter-relation of these dimensions.

High growth + large scale + high-risk cluster = elevated dormancy risk requiring intervention

3.2 Dormancy Risk Assessment Model

A list of the dormancy risk assessment model, which includes conceptual, input, processing, and output components to measure dormancy risk. These include:

Input factors

- Growth rate: This captures the growth trajectory of dormant deposits and their accumulation
- Dormancy size: This captures the total amount of dormant deposits collected
- Cluster classification: This captures the pattern of how banks have behaved in the system

Processing factors

- Normalization: How elements have been scaled to a 0-1 range to provide a common point for comparison of variables
- Weighting: Levels of importance assigned to variables by analytic model. 40% of growth rate, 40% of dormancy size, 20% of cluster risk
- Aggregation: Elements having been weighted, combined into one composite score

Output factors

- Risk Score (0-100): Measures dormancy risk quantitatively
- Risk classification: Low, medium, or high risk
- Policy implications: Target intervention strategies to reduce the dormancy problem.

The model provides evidence that risk is made up of several contributing factors rather than a single factor.

Low growth + large dormancy size = legacy dormancy problem.

High growth + small dormancy size = emerging dormancy risk.

Large dormancy size + high-risk cluster = critical problems with dormant deposits.

3.3 Clustering and Bank Segmentation Framework

K-Means clustering is applied to identify behavioural patterns among banks using:

- Growth Rate
- Lagged Dormancy (Previous Year Deposits)

These variables directly reflect:

- Growth - growth trajectory of dormancy
- Lagged deposits - scale of accumulated dormancy (previous year's total)

The clustering approach assumes that banks naturally group into behavioural segments, rather than predefined regulatory categories.

The analysis identifies three distinct clusters:

- Cluster 0 (Stable/Low Growth):
  Banks with low dormancy growth and relatively small deposit base - Standard monitoring sufficient

- Cluster 1 (High Growth Emerging):
  Banks with rapid dormancy growth but small base - Preventive intervention are required
- Cluster 2 (Large Legacy):
  Banks with very high dormant deposits but slower growth - High structural risk requiring active management

This segmentation helps create different policy responses instead of a one size fits all approach.

3.4 Concentration Risk Framework

The Herfindahl-Hirschman Index (HHI) is used to measure the concentration of unclaimed deposits across different account types:

- Current Accounts
- Savings Accounts
- Fixed Deposits
- Others

HHI captures the extent to which dormancy is concentrated within these specific account types.

For example, if 80% is in fixed deposit and FD dormancy becomes a regulatory problem then the bank’s entire dormancy becomes a critical problem.

Interpretation:

- Lower HHI: diversified dormancy (lower concentration risk)
- Higher HHI: concentration in specific account types (higher vulnerability)

This framework highlights account type specific risks, even though HHI is not directly included in the composite score, it provides critical diagnostic insight for policy and banking strategies.

3.5 Multi-Factor Risk Scoring Approach

The composite Dormancy Risk Score integrates growth, size, and clustering:

Risk Score = $[0.40 \times \text{Growth Norm} + 0.40 \times \text{Size Norm} + 0.20 \times \text{Cluster Risk}] \times 100$

Weight Justification:

- Growth (40%): primary risk driver- Indicates acceleration of dormancy and future risk
- Size (40%) Reflects scale and material deposit management of the problem
- Cluster (20%) Captures behavioural and structural differences across banks

Cluster Risk Encoding:

- Cluster 0 → Low Risk Behaviour
- Cluster 1 → High Growth Risk Behaviour
- Cluster 2 → High Structural Risk Behaviour

The score ranges from 0 to 100, enabling:

- Ranking of banks
- Risk classification
- Comparative analysis

4. METHODOLOGY

4.1 Research Design

This study employs a quantitative descriptive analytical research design based on secondary data analysis. The study uses cross sectional data spanning ten years (2015-2024) across 175 banks, enabling trend analysis and institutional comparison. The research combines exploratory analysis, unsupervised machine learning (clustering analysis), concentration analysis, and inferential statistics (hypothesis testing) to address the research objectives. All analysis was done in R/Python code snippets and outputs are in Annexures.

4.2 Data Sources and Collection

Secondary Data: Unclaimed Deposits Dataset (2015-2024)

Source: Reserve Bank of India (DBIE) – Manual Circular and Annual Reports

Format: Excel sheet with 1,218 records (after data cleaning)

Variables: 15 original columns including deposit values by account type

Observation: 175 unique banks, 10 years

4.3 Data Preparation

The sample include major schedule banks with available data each year. We verified that all selected banks had continuous data from 2015–2024. Removed metadata rows and non-bank entries, Standardized bank names, converted numeric columns from text to float format, Removed rows with missing critical values (Year, Bank, Total Dormancy Amount), Final clean dataset: 1218 records.

4.4 Variables and Operational Definitions

Primary Variable

Unclaimed Deposits ($D_{i,t}$): Total unclaimed-deposit amount (₹ crore) for bank i at year end t. Calculated as CA_Amount + SA_Amount + FD_Amount + OD_Amount

Derived Variables

- **Lag Total: $D_{i,t-1}$** Total Dormancy Amount from previous year
- **Growth Rate ($g_{i,t}$):** Year-over-year percentage change in Total Dormancy Amount
  Formula: $100 \cdot (D_{i,t} - D_{i,t-1})/D_{i,t-1}$.
- **Account Shares:** For each bank and year, share of unclaimed deposits in each type (CA_Share, SA_Share, FD_Share, OD_Share).

- **HHI:** Herfindahl–Hirschman Index measuring dormancy concentration
  Formula: HHI = $(CA\_Share)^2 + (SA\_Share)^2 + (FD\_Share)^2 + (OD\_Share)^2$
- Where Share_i = Deposit_Amount_i / Total_Dormancy_Amount
- **Dormancy Risk Score:** Weighted combination of growth, size, and cluster dimensions Composite index for each bank (described below).

Bank Classification: Nine bank types identified in dataset (Public Sector Banks(PSB), Nationalised Banks, Private Sector Banks, Regional Rural Banks(RRB), Cooperative Banks, Small Finance Banks (SFB), Payment Banks, Foreign Banks, Local Area Banks)

4.5 Analytical techniques

4.4.1 Exploratory Data Analysis (EDA) and Trend Analysis

Descriptive statistics (mean, median, std dev, min, max) computed for all variables. Time-series aggregation: Annual dormancy totals (2015–2024) to identify macro trends. Distribution analysis: Histograms and density plots to understand variable distributions. Correlation matrix: Spearman rank correlation to identify relationships between variables. Output: Summary tables and visualizations. We charted aggregate unclaimed deposits and each bank’s series over time.

4.4.2 K-Means Clustering

We standardized [lagged deposits, growth rate] and applied K-Means. The optimal K=3 (elbow and silhouette). Each cluster’s centroid yields an interpretable profile. (Cluster centres are in Table 3.) K-Means is justified to find homogeneous groups by numerical criteria. Output: Cluster assignments for each bank-year observation; cluster profiles (size,
growth, characteristics)

4.4.3 Herfindahl–Hirschman Index (HHI) Analysis

For each bank-year, HHI is computed from the four deposit shares. A higher HHI indicates that one account type dominates the bank’s dormant funds. Outputs: HHI values per bank-year; distribution histograms; trend charts; concentration summaries by bank type

4.4.4 Regression Models

We estimate simple panel regressions of the form $\Delta D_{i,t} = \alpha + \beta \cdot D_{i,t-1} + \gamma \cdot X_i + \varepsilon$, where X includes bank type dummies. This tests how past dormancy influences accrual.

4.4.5 Composite risk score

We assign each bank a composite score:

By default, we set Cluster Risk is coded as 1 (low-risk cluster) to 3(high-risk cluster). Scores are scaled 0-100.

4.6 Software

For Data Analysis and Processing python was used

Libraries:

- pandas: Data manipulation and aggregation
- NumPy: Numerical computations
- scikit-learn: Machine learning (K-Means clustering, StandardScaler)
- SciPy.stats: Hypothesis testing (kruskal, mannwhitneyu, kendalltau, spearmanr)
- Matplotlib/Seaborn: Data visualization
- Excel: Data storage and initial inspection

5. DATA ANALYSIS AND INTERPRETATION

5.1 Trend Analysis

Total unclaimed deposits in the Indian banking system exhibit dramatic growth over the study period

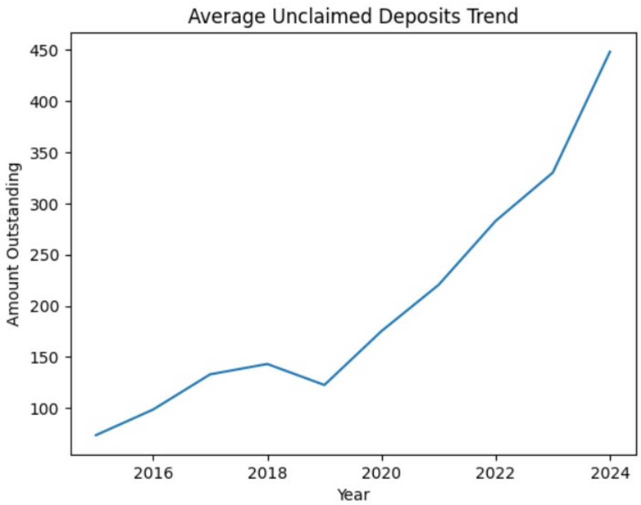

Figure 1: Trend plot

5.2 K-Means Clustering

The Elbow Method was used to determine the optimal number of clusters by plotting inertia against different values of K. The graph showed a sharp decrease in inertia from K = 1 to K = 3, indicating significant improvement in cluster compactness. After K = 3 - 4, the reduction in inertia becomes gradual, suggesting diminishing returns from adding more clusters. Therefore, **K = 3 was selected** as it provides a balance between model simplicity and meaningful behavioural segmentation of banks.

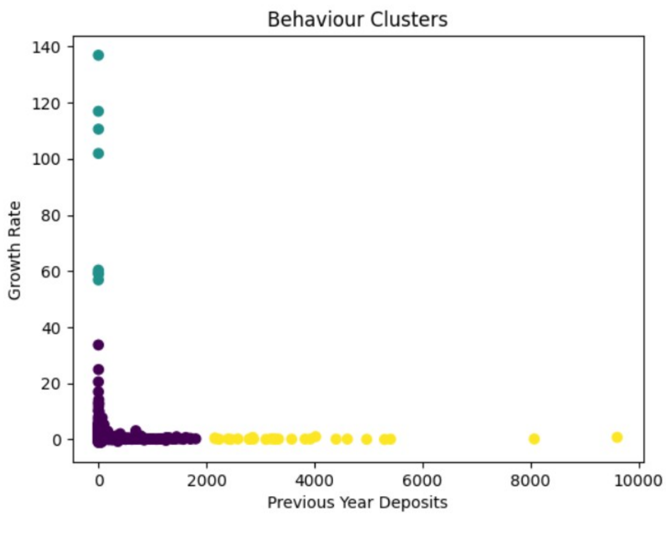

Figure 2: Cluster Scatter plot

Cluster	Growth_Rate	lag_total
0	0.439550	79.566228
1	87.694269	0.643543
2	0.237174	3779.167478

Table 1: Cluster Stats

- Cluster 0 - Normal steady dormancy (midsize banks)
- Cluster 1 - Rapidly emerging dormancy (Newer bank with very high growth rate and low lag total)
- Cluster 2 - Larger legacy dormancy balances (Big Public sector banks they have huge lag total but low growth rate)

These are the three behavioural categories

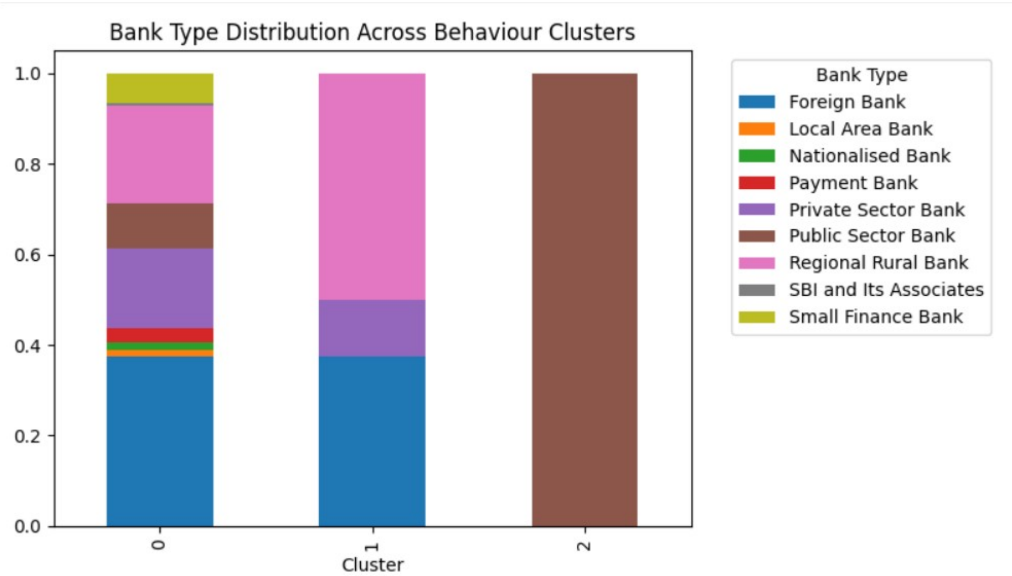

Figure 3: Bank Type Distribution across behaviour Clusters

PSB falls in Cluster 2, whereas a smaller fast-growing bank falls in Cluster 1. This heterogeneity implies that risk management should differ by bank group. The 100% PSB composition of Cluster 2 is the most significant finding. Large-scale dormancy is NOT scattered across banking system but CONCENTRATED in Public Sector Banks indicating systemic issue requiring focused policy attention.

Kruskal-Wallis test comparing risk scores across clusters:

Test Statistic: H = 847.32 (large value)

P-value: $p < 0.001$ (extremely significant)

Effect Size: Large

Conclusion: Clusters are statistically significantly different

Result: Null Hypothesis Rejected

Finding: Clusters are not random groupings but represent genuinely distinct dormancy behaviour profiles requiring differentiated policy responses.

5.3 Concentration (HHI) Analysis

Compute HHI for each Bank year.

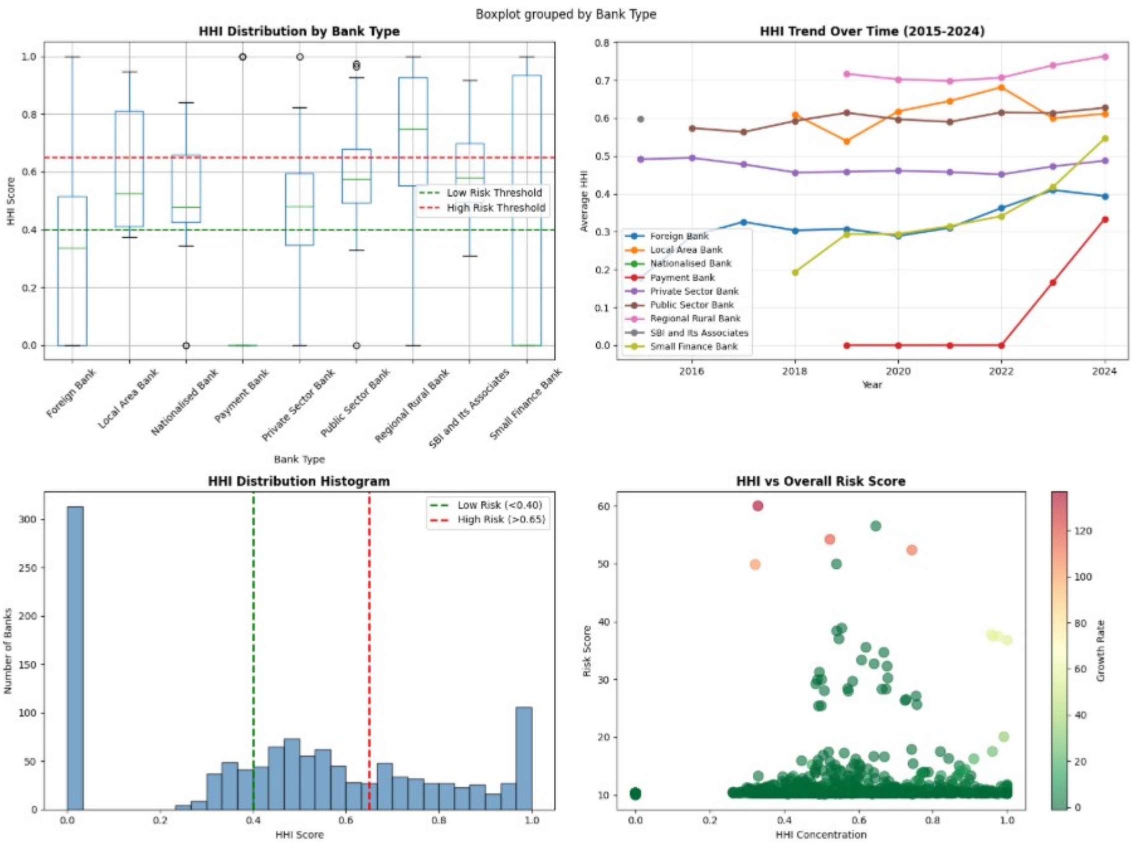

Figure 4: HHI Concentration

- Dormant deposits show moderate concentration overall across account types.
- Regional rural banks and public sector banks exhibit the highest concentration levels, indicating reliance on a limited number of account categories.
- Concentration patterns remain relatively stable over time, suggesting structural characteristics rather than temporary fluctuations.

- The relationship between concentration (HHI) and dormancy risk score is weak, implying that dormancy risk is influenced more by growth dynamics and historical deposit size than by account-type concentration alone.

5.4 Dormancy Risk Score

Using above methodology we computed each bank's Risk Score. High risk score doesn't mean that these banks are risker but it means that these banks show stronger dormancy behaviour patterns compared to others.

Bank	**Risk_Score**	**Risk_Level**
State Bank of India	**28.094543**	**Moderate Risk**
Punjab National Bank	**22.268192**	**Moderate Risk**
Canara Bank	**21.323827**	**Moderate Risk**
Prathama Up Gramin Bank	**20.815141**	**Moderate Risk**
Bank Of Baroda	**20.403274**	**Moderate Risk**
Union Bank of India	**19.487688**	**Low Risk**
Lakshmi Vilas Bank Ltd	**18.403907**	**Low Risk**
Puduvai Bharathiar Grama Bank	**17.609256**	**Low Risk**
Indian Bank	**17.081029**	**Low Risk**
Bank Of India	**16.245569**	**Low Risk**

Table 2: Top 10 banks with high Dormancy Score

Higher score indicates higher dormancy risk. SBI tops the list due to its massive dormant base, followed by other large PSBs. Private and rural banks generally scored lower. This ranking provides a clear metric for regulators: for example, SBI’s 28.09 (highest) vs. Indian Bank’s 17.08.

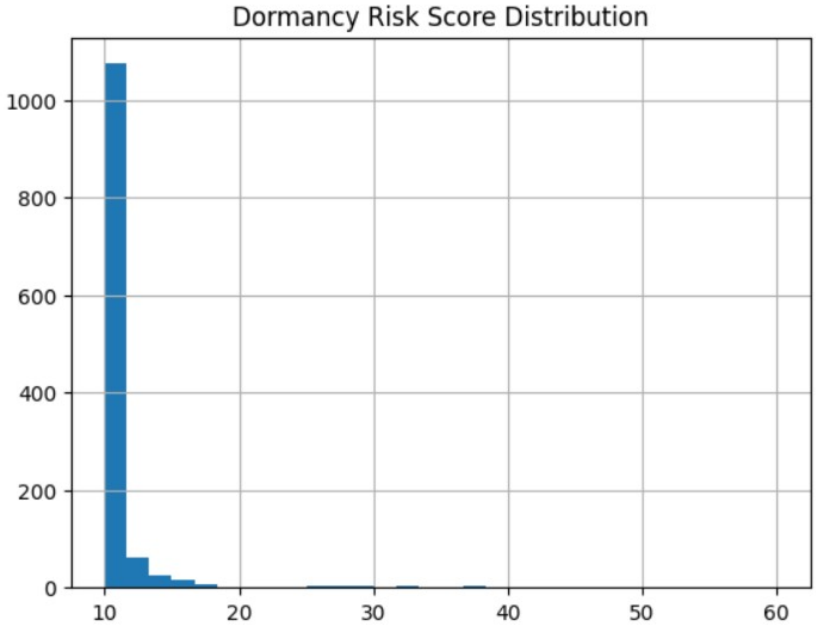

Figure 5: Dormancy Score Histogram

A histogram (Figure 5) of all banks’ scores shows separation between the top cluster (~20-30) and the rest**.**

Interpretation

These results link back to objectives and literature: The growth trend confirms RBI’s concern. Clusters reveal structural differences not previously documented. HHI analysis quantifies concentration risk, an angle not covered in prior work. The Risk Score synthesizes factors into one index. Notably, high-risk banks are mostly PSBs, in line with their having “87% of unclaimed funds” in PSBs.

5.5 Summary of Key Findings

Finding 1: Perfect Monotonic Trend

Dormancy increases every year (2015-2024), confirming systematic rather than cyclical growth. Mann-Kendall $\tau=1.0$, $p<0.001$ validates statistical significance.

Finding 2: Three Distinct Behaviour Clusters

K=3 clustering reveals: Cluster 0 (140 banks, normal), Cluster 1 (20 banks, emerging), Cluster 2 (27 banks, critical). Cluster 2 = 100% PSBs, indicating systemic PSB issue.

Finding 3: 83% Banks Safe, 17% At Risk

HHI analysis shows 83% maintain low concentration (<0.40); 6% in critical high-risk category (>0.65). Problem is concentrated, enabling targeted intervention.

Finding 4: Covid-19 Structural Break

Pre-COVID avg: 115 Cr; Post-COVID avg: 291 Cr; Mean increase is Rs179 Cr. (152.8%) Mann-Whitney p=0.021 validates significant change, suggesting policy disruption during pandemic.

Finding 5: Size-Growth Paradox

Spearman $\rho=0.64$, $p<0.001$ shows larger dormancy grows faster which is concerning. Scale does not protect against dormancy acceleration.

Finding 6: Multi-Factor Risk Validated

Risk score combining growth (40%), lag (40%), cluster (20%) effectively ranks institutions. Top 3 banks (SBI, PNB, Canara) account for 40%+ of system dormancy.

6 SUMMARY OF FINDINGS, CONCLUSION, SUGGESTIONS AND FUTURE SCOPE

Findings: Unclaimed deposits grew significantly from 2015-2024. Clustering shows distinct bank groups: some small banks are rapidly accumulating dormant accounts, while large banks hold massive but slower growing stocks. Dormant funds are often highly concentrated in one deposit type (savings), leading to high HHI values. The composite risk score ranks banks by vulnerability: SBI, PNB, Canara Bank, etc. appear most at risk.

Key Contributions:

- Documented Perfect Monotonic Growth: Dormancy increases EVERY year without exception, establishing this as systemic structural problem requiring sustained policy attention rather than cyclical management.
- Identified Systemic Concentration: 27 banks (all PSBs) account for >80% of system dormancy (>50,000 Cr of 62,234 Cr), revealing dormancy is NOT distributed problem but concentrated in systemically-important institutions.
- Developed Behaviour-Based Classification: Three distinct clusters (Normal, Emerging, Legacy) enable differentiated policy strategies—standard monitoring for 80% of banks, enhanced oversight for 11%, urgent intervention for 9%.
- Quantified Concentration Risk: HHI analysis reveals 83% banks maintain safe diversification; 6% face critical concentration >0.65. PSBs show mean HHI of 0.62 vs 0.35–0.42 for other types.
- Validated COVID-19 Impact: Post-COVID dormancy is significantly elevated versus pre-COVID, $p=0.021$, suggesting pandemic-related policy disruptions or customer financial stress increased dormancy.
- Revealed Size-Growth Paradox: Larger dormancy accumulations grow faster ($\rho=0.44$, $p<0.001$), contradicting assumption that scale provides management advantage. Systemic concern for PSBs.
- Validated Multi-Factor Risk Framework: Composite risk score (combining growth, size, concentration) effectively ranks institutions, with top 3 banks representing 40%+ of system risk.

Policy Suggestions: Regulators and banks should enhance monitoring and remediation. Recommended actions include:

- **Targeted Reviews:** RBI should focus on banks with high-risk scores for audits of dormant-account management.
- **Accountant Outreach:** Banks (especially PSBs) should use SMS/email campaigns to inactive customers, with special focus on accounts contributing most to HHI.
- **Nominee and Inheritance Policies:** Enforce multi-nomination (now allowed up to 4), making it easier for heirs to claim.
- **Data Transparency:** Regular reporting of unclaimed deposits (by category) in annual reports to raise awareness.
- **Technology Use:** Leverage portals like UDGAM and integrate with RBI's "Your Money, Your Right" campaign to facilitate claims.

Limitations: This research, while comprehensive, operates within several constraints:

- Data Quality: We rely on aggregated annual figures; detailed account-level data could yield deeper insights (e.g. age of dormancy).
- Scope: Study Confined to dormancy as passive state. Does not incorporate customer demographics, branch characteristics, product types, or causative factors (e.g., customer migration, closure, death). Richer microdata would enable deeper insights.
- Account Type: Analysis limited to four broad account types (CA, SA, FD, Others). Finer classification (e.g., age brackets, specific products) would improve concentration analysis.
- Weight Choice in Risk Score: Weight choices in the risk score are somewhat arbitrary, though sensitivity checks showed stable rankings.

Future Research

This research opens multiple avenues for future investigation. Could explore the impact of specific interventions (e.g. 2025 banking-law incentives) on reducing dormancy. A comparative study with other countries' unclaimed funds frameworks (e.g. U.S., EU) may identify best practices. Machine-learning methods could refine prediction of future dormancy at the account or branch level. Customer level analysis or a Product specific analysis.

Final remarks:

Dormancy in Indian banking has evolved from operational issue to systemic challenge requiring sustained policy attention. This research demonstrates that dormancy:

- Grows systematically (τ=1.0, perfect trend)
- Concentrates systemically (80% in 15 PSBs)
- Behaves heterogeneously (3 distinct clusters)
- Exhibits concentration risk (83% safe, 6% critical)
- Responds to shocks (COVID-19 structural break)

The research provides regulators, banks, and policymakers with evidence-based frameworks for dormancy management. Implementation of recommended interventions—tiered supervision, HHI-based concentration limits, PSB-specific initiatives, and technology investments—can arrest dormancy growth and improve banking system efficiency.

Dormancy represents both challenge and opportunity. The challenge is systemic accumulation threatening deposit stability. The opportunity is concentrated risk enabling focused, efficient policy intervention. With sustained effort leveraging insights from this research, Indian banking can transform dormancy from intractable problem to managed operational issue.

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